

Deep Learning — Tutorial 1

Prerequisites: Goodfellow, Bengio & Courville, Chapters 2–5

Practice set — not for submission

These questions are for your own practice and do not need to be submitted or graded. Work them through properly and write out each derivation in full rather than reading the question and deciding you could probably do it. The gap between those two states is where most of the learning happens.

Q1. Linear algebra warm-up

- (a) For $x = [3, -4, 0, 1]^\top$, compute $\|x\|_1$, $\|x\|_2^2$, and $\|x\|_\infty$.
- (b) Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Find its eigenvalues and state whether A is positive definite, justifying your answer in one line.

Q2. Pseudoinverse

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$.

- (a) Write down the singular values of A and compute A^+ .
- (b) Verify that $A^+A = I_2$ but $AA^+ \neq I_3$. Explain in one sentence what this tells you about solving $Ax = y$.

Q3. Information theory

Let $P = \text{Bernoulli}(0.5)$ and $Q = \text{Bernoulli}(0.9)$.

- (a) Compute $H(P)$ in nats.
- (b) Compute $D_{\text{KL}}(P \parallel Q)$ and $D_{\text{KL}}(Q \parallel P)$ in bits.
- (c) Your answers in (b) differ. If Q is a model being fitted to data distributed as P , explain which direction of the KL divergence you would minimise and why.

Q4. Numerical stability of softmax

- (a) Prove that $\text{softmax}(x) = \text{softmax}(x + c\mathbf{1})$ for any scalar c .
- (b) Compute $\text{softmax}([1000, 1001, 1002])$ using the stabilisation trick, and state what a naive floating-point implementation would return instead.
- (c) Even with a numerically stable `softmax`, computing $\log(\text{softmax}(x))$ can fail. Explain why, and write down the correct expression for `log_softmax`.

Q5. Maximum likelihood and mean squared error

Assume the model $p(y \mid x; w) = \mathcal{N}(y; \hat{y}(x; w), \sigma^2)$ with σ^2 fixed and known, and an i.i.d. training set $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$.

- (a) Write the conditional log-likelihood $\sum_i \log p(y^{(i)} \mid x^{(i)}; w)$ in full.
- (b) Show that maximising it with respect to w is equivalent to minimising the mean squared error.
- (c) State precisely where the assumption of constant σ^2 is used, and what would change if σ^2 depended on x .

Q6. Short conceptual answers

Answer each in **two or three sentences only**.

- (a) Distinguish *representational capacity* from *effective capacity*, and name one cause of the gap between them.
- (b) A classmate says: “The no free lunch theorem proves that no learning algorithm is better than any other, so benchmarking is pointless.” State what is wrong with this.
- (c) A grid with 10 divisions per axis in d dimensions requires at least 10 examples per cell. How many examples are needed at $d = 3$ and $d = 10$? What does this imply about relying on a local-constancy (smoothness) prior in high dimensions?